Department of Computer Science & Engineering

PROJECT FILE

ON

HOSTEL ELECTRICITY USAGE PREDICTOR



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CERTIFICATE

I hereby certify that the work which is being presented in the B.Tech. Major Project Report entitled **"HOSTEL ELECTRICITY USAGE PREDICTOR",** in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Science & Engineering** and submitted to the Department of Computer Science & Engineering of CGC-College of Engineering Landran, Mohali Punjab is an authentic record of my own work carried out during a period (3rd **semester**) under the supervision of **Dr. Seema Rani (Assistant Professor) , Department of CSE**.

The matter presented in this Project Report has not been submitted by me for the award of any other degree elsewhere.

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This is to certify that the above statement made by the student(s) is correct to the best of my knowledge.

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ACKNOWLEDGEMENT

I would like to place on record my deep sense of gratitude to Dr. Sushil Kamboj, HOD-Dept. **Computer Science & Engineering**, CGC- College of Engineering Landran, Mohali, India for his generous guidance, help and useful suggestions.

I express my sincere gratitude to Dr. Seema Rani , Dept. **Computer Science & Engineering**, CGC- College of Engineering Landran, Mohali, India, for his stimulating guidance, continuous encouragement and supervision throughout the course of present work.

I am extremely thankful to Dr. Anuj Kumar Gupta , Principal, CGC- College of Engineering Landran, Mohali, for providing me infrastructural facilities to work in, without which this work would not have been possible.

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ABSTRACT

Electricity is an important resource in hostel life, where students use different electrical appliances such as fans, lights, laptops, and other devices every day. Excessive or unnecessary use of electricity can lead to higher energy consumption and increased electricity costs. Therefore, estimating electricity usage can help students understand their consumption patterns and encourage them to use electricity more efficiently.

The **Hostel Electricity Usage Predictor** is a machine learning-based project developed to predict the daily electricity consumption of a hostel room. The main objective of this project is to estimate electricity usage based on the number of electrical appliances, their daily usage hours, and the number of occupants in a room. The project is developed using Python and the Multiple Linear Regression algorithm, which is used to predict continuous numerical values.

The system uses a dataset containing information about appliance usage and estimated electricity consumption. The dataset is processed using Python libraries such as Pandas and NumPy, while Scikit-learn is used to train and evaluate the machine learning model.
Matplotlib is used to represent the data graphically, and Streamlit is used to develop an interactive user interface through which users can enter room details and view the predicted electricity consumption in kilowatt-hours (kWh).

In the current implementation, a simulated dataset is used to demonstrate the working of the system and the complete machine learning process. The model is trained using a portion of the dataset and evaluated using the remaining data. Performance measures such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) are used to evaluate its predictions.

This project demonstrates how machine learning and data visualization can be applied to a practical problem related to electricity management in hostels. In the future, the system can be improved by collecting actual hostel electricity consumption data, including more electrical appliances, comparing different machine learning algorithms, and integrating smart meters to improve prediction accuracy.

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ABBREVIATIONS AND NOMENCLATURE

Term	Meaning
AI	Artificial Intelligence
CSV	Comma-Separated Values
kWh	Kilowatt-hour; a unit of energy
MAE	Mean Absolute Error
ML	Machine Learning
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
UI	User Interface
W	Watt; a unit of power
X	Input feature matrix
y	Target variable
$\hat{y}$	Predicted target value

CHAPTER 1: INTRODUCTION

1.1 General

Electricity consumption is a practical concern in student hostels because multiple occupants use electrical appliances within a shared room. The amount of energy used depends on the number and type of appliances, their power ratings, the duration for which they operate, and the behaviour of the occupants. A simple way to understand this relationship is to record the inputs and estimate energy consumption in kilowatt-hours.

The Hostel Electricity Usage Predictor is designed as an educational application that combines basic Python programming with a supervised machine learning algorithm. It allows a user to enter information about a room and obtain a model-generated estimate. The application also presents visualizations so that users can inspect the sample data and understand the difference between predicted and target values.

1.2 Background

Traditional electricity bills generally report consumption over a billing period. Such totals are useful for accounting, but they may not directly show how daily usage relates to individual appliances in a particular room. An estimate based on appliance usage can provide an approachable demonstration of energy calculations and data-driven prediction. However, the quality of any estimate depends on the reliability of input information, the assumed or measured appliance power, and the data used to train the model.

Machine learning offers methods for learning patterns from examples. In supervised learning, each training record contains input features and a known target value. A regression algorithm is suitable when the target is numerical, such as energy consumption. Multiple Linear Regression is an appropriate first model for a student project because its general idea is understandable and it is available in standard machine learning libraries.

1.3 Motivation

The motivation for this project is to connect classroom concepts with a familiar setting. Hostel rooms are easy for students to understand, and appliances such as fans, lights, and laptops can be represented as structured input data. The project gives students an opportunity to practise data preparation, feature selection, model training, evaluation, visualization, and development of a small interactive application.

Another motivation is to distinguish energy from power. Power is commonly measured in watts, while energy consumption is measured in kilowatt-hours. Explaining this distinction in the application helps avoid a common misunderstanding when discussing electricity use.

1.4 Aim of the Project

The aim is to develop a simple interactive machine learning prototype that estimates daily electricity consumption for a hostel room from appliance counts, operating hours, and the number of occupants, while clearly communicating the assumptions and limitations of the model.

1.5 Objectives

To develop a machine learning-based system that predicts daily hostel electricity consumption using appliance usage and room occupancy data, with interactive visualization and model evaluation.

1.6 Scope

The current scope is limited to a small software prototype. It accepts counts for fans, lights, and laptops, daily operating hours for these appliances, and the number of occupants. The prototype uses illustrative power ratings and a simulated dataset. It does not connect to a smart meter, automatically detect appliances, control electrical devices, or provide a verified bill forecast.

The application can be extended after data collection. A future version may accept actual meter readings, add appliances such as heaters or kettles, incorporate seasonal or day-of-week information, and compare several models. These are future possibilities and are not claimed as completed features.

1.7 Significance

The project is significant as a learning exercise because it integrates several commonly used tools in a single workflow. It provides a concrete example of regression, demonstrates how features and targets are represented, and shows how a model can be exposed through a simple

user interface. Its practical theme also gives the project a clear problem statement for a report and demonstration.

1.8 Organization of the Report

This project report is divided into five chapters, each covering a different aspect of the Hostel Electricity Usage Predictor.

Chapter 1: Introduction introduces the project, explains the problem, discusses the motivation behind the project, and presents its objectives and scope.

Chapter 2: Literature Survey provides an overview of the concepts, technologies, and tools related to electricity consumption prediction and machine learning. It also discusses the use of Multiple Linear Regression and data visualization in the project.

Chapter 3: Present Work and Implementation Methodology explains the problem statement, the proposed solution, the dataset, the tools and technologies used, and the steps involved in developing and training the machine learning model.

Chapter 4: Results and Discussion describes how to execute the application and explains its working through the user interface and graphical representations. It also includes the model evaluation process using performance metrics such as MAE, RMSE, and R^2. Actual screenshots and evaluation results will be added after running the application.

Chapter 5: Conclusion and Future Scope summarizes the work completed, highlights the usefulness of the project, discusses its current limitations, and suggests possible improvements for future development.

Finally, the **Appendix** contains the source code used to develop the prototype, along with any additional technical information required to understand the implementation.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

A literature survey helps in understanding the basic concepts, methods, and technologies related to a project. For the Hostel Electricity Usage Predictor, the main areas of study include electricity measurement, energy estimation, machine learning, regression, model evaluation, data visualization, and application development.

This chapter explains the concepts used in the project and describes how they support the development of an electricity consumption prediction system. It provides a conceptual overview of the selected methods and tools rather than claiming that a formal systematic review of hostel-specific research has been conducted.

2.2 Electricity Measurement and Energy Estimation

Electrical power refers to the rate at which an electrical device consumes energy. It is generally measured in watts (W). Electrical energy represents the amount of electricity consumed over a period of time and is commonly measured in watt-hours (Wh) or kilowatt-hours (kWh).

When the power rating and operating time of an appliance are known, its approximate energy consumption can be calculated by multiplying its power rating by the number of hours it operates. For multiple identical appliances, the estimated consumption is also multiplied by the number of appliances.

For example, if a fan has a power rating of 60 watts and operates for 10 hours, its estimated energy consumption is 600 watt-hours, or 0.6 kWh.

Although this calculation provides a useful estimate, actual electricity consumption may vary depending on the operating conditions of the appliance, changes in power usage, user behaviour, and the presence of other electrical devices. Therefore, actual electricity meter readings are more suitable for developing and validating a real-world prediction model.

2.3 Rule-Based Estimation

Rule-based estimation is a method of calculating electricity consumption using predefined rules, appliance power ratings, and operating hours. It provides a simple way to estimate how much energy different appliances may consume during a day.

The main advantage of this approach is that it is easy to understand and implement. Each appliance contributes an estimated amount of energy based on its power rating, quantity, and duration of use.

However, rule-based estimation depends on the accuracy of the assumptions used in the calculation. It may not fully represent changes in appliance usage, differences in actual power consumption, or electricity used by devices that have not been included in the calculation.

In the Hostel Electricity Usage Predictor, rule-based calculations provide a useful reference for understanding electricity consumption. They can also serve as a baseline for comparison when the machine learning model is evaluated using actual measured data.

2.4 Regression-Based Prediction

Regression is a supervised machine learning technique used to predict numerical values. It identifies relationships between input variables and a continuous output based on examples provided during training.

Multiple Linear Regression is one of the simplest and most widely used regression algorithms. It estimates the relationship between the input features and the target variable by fitting a Multiple linear equation to the training data.

In this project, the input features include the number of fans, lights, and laptops, their daily usage hours, and the number of occupants in a hostel room. The target variable is the estimated daily electricity consumption measured in kilowatt-hours (kWh).

After training, the model uses the learned relationships to predict electricity consumption for new combinations of input values. Multiple Linear Regression is selected because it is relatively simple to implement, computationally efficient, and suitable for demonstrating the basic working of a supervised machine learning system.

However, the model does not automatically establish cause-and-effect relationships. Its predictions depend on the quality and characteristics of the training data. Since the current prototype uses simulated data, its performance cannot be considered proof of accuracy on actual hostel electricity consumption.

2.5 Supervised Learning Workflow

Supervised learning involves training a model using data that contains input features and corresponding target values. The model learns patterns from the training examples and uses them to make predictions for new inputs.

The general supervised learning workflow followed in this project consists of the following steps:

1. **Problem Definition:** Identifying the need to estimate daily electricity consumption in hostel rooms.
2. **Dataset Preparation:** Creating a dataset containing appliance information, usage hours, occupant count, and estimated electricity consumption.
3. **Data Processing:** Organizing the data into a suitable format for machine learning.
4. **Feature and Target Selection:** Selecting appliance-related information as input features and electricity consumption as the target variable.
5. **Train-Test Splitting:** Dividing the dataset into training and testing subsets.
6. **Model Training:** Training the Multiple Linear Regression algorithm using the training data.
7. **Prediction:** Using the trained model to estimate electricity consumption for given inputs.
8. **Model Evaluation:** Measuring prediction errors and examining the model's performance using suitable evaluation metrics.
9. **Application Development:** Connecting the trained model to an interactive interface developed using Streamlit.

Separating training and testing data helps evaluate the model on examples that were not used during training. However, the reliability of this evaluation depends on how closely the dataset represents the real situation in which the model is intended to be used.

2.6 Evaluation Metrics

Evaluation metrics are used to measure the difference between the actual target values and the values predicted by a machine learning model. In this project, three metrics are considered: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Mean Absolute Error (MAE): MAE calculates the average absolute difference between the actual values and the predicted values. A lower MAE indicates that the predictions are closer to the target values on average.

Root Mean Squared Error (RMSE): RMSE calculates the square root of the average squared prediction errors. It gives relatively greater importance to larger errors and helps identify cases where the model makes predictions that differ significantly from the target values.

Coefficient of Determination (R^2): R^2 measures how well the model's predictions fit the target values compared with a baseline that predicts the mean target value. A higher R^2 can indicate a better fit, although its interpretation depends on the dataset and evaluation conditions.

When the target variable is daily electricity consumption, MAE and RMSE are expressed in kWh per day. These metrics help assess the model's performance, but results obtained from simulated data should be interpreted as performance on that simulated dataset rather than as proof of real-world accuracy.

2.7 Data Visualization

Data visualization is the process of presenting information through graphical representations. It helps users understand patterns, distributions, and relationships that may be difficult to identify from numerical data alone.

In the Hostel Electricity Usage Predictor, Matplotlib is used to create graphical representations of the dataset and the model's predictions.

A histogram is used to display the distribution of estimated electricity consumption values in the dataset. It helps show how the consumption values are spread across different ranges.

A scatter plot comparing target values with predicted values is used to examine the relationship between the model's predictions and the corresponding target values. Predictions closer to the ideal diagonal relationship indicate closer agreement between the predicted and target values.

These visualizations make the project easier to understand and help identify patterns that may require further investigation. However, graphical representations should be considered

alongside numerical evaluation metrics rather than being used as the only measure of model performance.

2.8 Python and Supporting Libraries

Python is used as the main programming language for developing the Hostel Electricity Usage Predictor. Several libraries support data preparation, model training, evaluation, visualization, and application development.

Tool / Library	Role in the project
Python	Main programming language.
Pandas	Creates and manipulates tabular data.
NumPy	Generates numerical arrays and random variation.
Scikit-learn	Splits data, fits Multiple Linear Regression, and calculates metrics.
Matplotlib	Creates the histogram and target-versus-prediction plot.
Streamlit	Provides an interactive browser-based interface.

The use of these tools simplifies the development process and allows different stages of the machine learning workflow to be combined into a single application.

2.9 Research Gap and Project Positioning

Estimating electricity consumption requires information about the appliances being used, their operating duration, and the conditions under which they operate. Simple calculations based on appliance ratings can provide approximate estimates, while machine learning techniques can be used to learn relationships between input features and consumption values from a dataset.

The Hostel Electricity Usage Predictor is developed as an educational prototype that combines electricity consumption estimation, Multiple Linear Regression, data visualization, and an interactive interface. Its purpose is to demonstrate how machine learning techniques can be applied to an everyday energy-management problem.

The current implementation uses a simulated dataset rather than actual hostel electricity meter readings. Therefore, the project does not claim to provide validated real-world predictions. A useful direction for further development would be to collect daily electricity readings from hostel rooms, compare predicted consumption with measured consumption, and evaluate the model using real data. This would provide a stronger basis for assessing its practical usefulness.

2.10 Summary

This chapter discussed the main concepts and technologies related to the Hostel Electricity Usage Predictor. It covered electricity measurement, rule-based estimation, regression-based prediction, supervised learning, evaluation metrics, data visualization, and the Python libraries used in the project.

Based on these concepts, Multiple Linear Regression has been selected as the initial prediction algorithm, while Pandas, NumPy, Scikit-learn, Matplotlib, and Streamlit support the different stages of development. The project provides a practical demonstration of a machine learning workflow. However, its real-world reliability can only be established through testing and evaluation using actual hostel electricity consumption data.

CHAPTER 3: PROBLEM STATEMENT AND IMPLEMENTATION METHODOLOGY

3.1 Problem Statement

Electricity is used regularly in hostels for operating appliances such as fans, lights, laptops, and other electrical devices. The amount of electricity consumed varies from room to room depending on the number of appliances, their daily usage hours, and the number of occupants. However, students may not always be aware of how their daily activities affect electricity consumption.

The main problem addressed by this project is to develop a simple application that can estimate the daily electricity consumption of a hostel room based on selected appliance and room details. The system uses machine learning to generate an estimated consumption value in kilowatt-hours (kWh).

The initial version of the project uses a simulated dataset to demonstrate the working of the machine learning model. It is intended for educational purposes and does not replace an electricity meter or provide verified measurements of actual hostel electricity consumption.

3.2 Proposed Solution

The proposed solution is a Python-based application that uses the Multiple Linear Regression algorithm to estimate the daily electricity consumption of a hostel room. The user provides information such as the number of fans, lights, and laptops, their daily usage hours, and the number of occupants.

The application generates or loads a dataset, prepares the input features and target variable, trains the machine learning model, and uses the trained model to produce predictions.
Streamlit provides an interactive interface through which users can enter room details and view the estimated electricity consumption.

The project is divided into two main components. The first component generates the simulated dataset using predefined assumptions, while the second component trains the model, accepts user inputs, displays predictions, and presents evaluation metrics and graphs. This separation makes the project easier to understand, test, and maintain.

3.3 Functional Requirements

Functional requirements describe the main operations that the system should perform. The functional requirements of the Hostel Electricity Usage Predictor are as follows:

1. The system should generate reproducible sample records for demonstrating the project.
2. It should represent appliance counts, daily usage hours, and the number of occupants as input features.
3. It should use daily electricity consumption in kWh as the target variable.
4. It should divide the dataset into training and testing subsets.
5. It should train a Multiple Linear Regression model using the training data.
6. It should provide an interactive interface for entering room and appliance details.
7. It should display the predicted daily electricity consumption.
8. It should calculate an estimated monthly consumption by multiplying the daily prediction by 30.
9. It should display a histogram and a scatter plot comparing target values with predicted values.
10. It should display evaluation metrics such as MAE, RMSE, and R^2.
11. It should clearly communicate that the dataset is simulated and that the results have not been validated using actual hostel meter readings.

3.4 Non-Functional Requirements

Non-functional requirements describe the quality, usability, and reliability expected from the application.

Usability: The application should have a simple interface with clearly labelled input fields and appropriate units so that users can enter information easily.

Reproducibility: The dataset generator should use a fixed random seed so that the same sample data can be generated again when the code and software environment remain unchanged.

Maintainability: The code should be organized into understandable files and functions so that modifications and improvements can be made easily.

Transparency: The application should clearly display its assumptions, evaluation metrics, and limitations. Users should understand that the predictions are estimates rather than actual meter readings.

Portability: The project should run on a compatible Python environment with the required libraries installed. It can be developed and executed on operating systems such as Windows, macOS, and Linux.

3.5 Input Features and Target Variable

The machine learning model uses information about the appliances and occupants of a hostel room to estimate daily electricity consumption.

Feature	Description	Unit / Type
fans	Count of fans in the room	Integer count
lights	Count of lights in the room	Integer count
laptops	Count of laptops used	Integer count
fan_hours_per_day	Daily operating duration for fans	Hours/day
light_hours_per_day	Daily operating duration for lights	Hours/day
laptop_hours_per_day	Daily operating duration for laptops	Hours/day
occupants	Number of people in the room	Integer count
daily_consumption_kwh	Target energy consumption	kWh/day

The first seven variables are used as input features, while daily_consumption_kwh is the target variable that the model learns to predict.

3.6 Dataset Design

The initial version of the project uses a simulated dataset containing 1,200 examples. The dataset is generated using Python and a fixed random seed so that the same data can be reproduced under the same conditions.

The dataset contains information about the number of fans, lights, and laptops, their daily operating hours, and the number of occupants in a room. These values are selected from predefined ranges to represent different possible room conditions.

For the initial demonstration, the dataset generator uses illustrative power ratings of 60 watts per fan, 10 watts per light, and 60 watts per laptop. It also includes an occupant-related term and random variation to represent additional loads and differences that are not directly described by the selected features.

These values are assumptions used to create example data and are not measured specifications for every appliance found in a hostel. Actual power consumption may differ according to appliance models, operating modes, and usage patterns.

The purpose of this dataset is to demonstrate the complete machine learning workflow, including data preparation, model training, prediction, and evaluation. The generated data should not be treated as a verified record of actual hostel electricity consumption.

3.7 Energy Unit Calculation

Electrical energy consumption can be estimated using the power rating of an appliance, the number of appliances, and their operating time.

The general formula is:

Energy Consumption (kWh) = (Power in Watts × Number of Appliances × Operating Hours) / 1000

For example, consider two fans, each having an assumed power rating of 60 watts, operating for 10 hours per day.

Energy consumption = (60 × 2 × 10) / 1000 Energy consumption = 1.2 kWh per day.

This calculation assumes that both fans continuously consume their rated power throughout the operating period. Actual consumption may vary depending on the type of fan and its operating conditions.

The example is a direct calculation based on an assumed power rating. It is not a prediction produced by the machine learning model.

3.8 Data Preparation

Data preparation is an important step in the machine learning process because the dataset must be arranged in a suitable format before the model can be trained.

In this project, the dataset is represented using a Pandas DataFrame. The input features are selected and stored in a variable named X, while the target variable, daily electricity consumption, is stored in a variable named y.

The dataset is then divided into training and testing subsets using the train_test_split function from Scikit-learn. Approximately 80% of the examples are used for training, while the remaining 20% are used for testing.

A fixed random state is used to make the division reproducible. This allows the model evaluation process to be repeated using the same generated dataset and split, provided the code and software environment remain unchanged.

Since both subsets are obtained from the simulated dataset, the resulting evaluation measures the model's performance on the generated examples rather than establishing its accuracy on real hostel data.

3.9 Multiple Linear Regression Algorithm

Multiple Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values. It identifies a linear relationship between the input features and the target variable by estimating suitable coefficients from the training data.

For a model with multiple input features, the general equation can be written as:

[
\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \cdots + \beta_px_p
]

Where:

- (\hat{y}) represents the predicted daily electricity consumption.
- (\beta_0) represents the intercept.

- (\beta_1, \beta_2, \ldots, \beta_p) represent the coefficients learned by the model.
- (x_1, x_2, \ldots, x_p) represent the input features.

In this project, the input features include appliance counts, daily operating hours, and the number of occupants. The output is the estimated daily electricity consumption in kWh.

The model is implemented using the Multiple Linear Regression class provided by Scikit-learn. During training, the algorithm estimates the coefficients that best fit the training examples according to the Multiple Linear Regression method.

Multiple Linear Regression has been selected because it is easy to implement, relatively simple to explain, and suitable for demonstrating a basic regression-based prediction system. However, it may not capture every complex or non-linear pattern that occurs in real electricity consumption.

3.10 Model Training and Testing

Model training is the process in which the algorithm learns relationships between the input features and the target variable. In this project, the Multiple Linear Regression model is trained using the training subset of the simulated dataset.

The training features are stored in X_train, and their corresponding target values are stored in y_train. The model learns from these examples using the fit() method.

After training, the model generates predictions for the testing features stored in X_test. These predictions are compared with the corresponding target values in y_test to evaluate performance.

The testing subset is kept separate from the model-fitting process so that the evaluation is performed on examples that were not used to train the model. However, because the training and testing subsets both originate from the same simulated data-generation process, the evaluation results only indicate how well the model fits that process.

3.11 Evaluation Metrics

Evaluation metrics help measure the difference between the target values and the values predicted by the model. The project uses three metrics: Mean Absolute Error, Root Mean Squared Error, and the coefficient of determination.

Mean Absolute Error (MAE): MAE calculates the average absolute difference between the target and predicted values. A lower MAE indicates a smaller average prediction error.

Root Mean Squared Error (RMSE): RMSE calculates the square root of the average squared prediction errors. It gives greater weight to larger errors and helps identify cases where predictions differ significantly from target values.

Coefficient of Determination (R^2): R^2 compares the model's prediction errors with the errors of a baseline that predicts the mean target value. It provides an indication of how well the model fits the evaluated data.

The application calculates and displays these metrics when it is executed. The actual numerical results will be recorded in the report after running the application. No metric values are assumed or inserted without obtaining them from an actual execution.

It is also important to note that these metrics are calculated using the simulated dataset. They should not be interpreted as evidence of the model's accuracy on real hostel electricity consumption.

3.12 System Architecture

The system follows a sequence of stages, beginning with data generation and input collection and ending with prediction, evaluation, and visualization.

Stage	Description	Output
User input	User enters appliance counts and usage hours.	Feature values
Data generator	Creates reproducible synthetic examples.	DataFrame
Preprocessing	Selects feature columns and target column.	X and y
Data split	Separates training and test examples.	Train/test subsets
Model training	Fits Multiple Linear Regression on training data.	Trained model

Prediction	Applies model to current user input.	Estimated kWh/day
Evaluation	Calculates MAE, RMSE, and R^2 on held-out examples.	Metrics
Visualization	Displays histogram and target-versus-prediction plot.	Charts in dashboard

The overall process connects data preparation, model training, prediction, and visualization in a single application. Streamlit provides the interface through which the user interacts with the system.

3.13 Implementation Environment

The project is developed using Python and its supporting libraries. The following table lists the main software and tools required to run the application.

Component	Requirement
Operating system	Windows, macOS, or Linux with a supported Python installation
Programming language	Python 3.10 or newer recommended
Editor	Visual Studio Code or another Python editor
Interface	Streamlit local web application
Libraries	Streamlit, Pandas, NumPy, Scikit-learn, Matplotlib

The required libraries are listed in the project's requirements.txt file. They can be installed together before running the application.

3.14 Steps to Run the Project

The following steps explain how to execute the Hostel Electricity Usage Predictor on a computer.

1. Extract the project ZIP file into a suitable folder.
2. Open the extracted project folder in Visual Studio Code or another compatible editor.
3. Open a terminal in the project directory.
4. Install the required libraries by executing the following command:

python -m pip install -r requirements.txt

5. After installation is complete, start the application using the following command:

python -m streamlit run app.py

6. Open the local URL displayed in the terminal. Streamlit generally provides a local address for accessing the application in a web browser.
7. Enter the required room details, including the number of appliances, their daily operating hours, and the number of occupants.
8. View the estimated daily electricity consumption, monthly extrapolation, graphical representations, and model evaluation metrics.
9. Record the evaluation metrics displayed by the executed program in the results section.

3.15 Default Demonstration Inputs

The application includes default input values to make the initial demonstration easier. These values can be changed by the user to represent a different hostel room.

Input	Default value	Status
Fans	2	Specified by project requester
Lights	3	Specified by project requester
Fan usage	10 hours/day	Specified by project requester
Laptop usage	5 hours/day	Specified by project requester
Occupants	4	Specified by project requester
Laptops count	1	Editable default assumption
Light usage	6 hours/day	Editable default assumption

The number of fans, lights, fan usage hours, laptop usage hours, and occupants follow the specified demonstration inputs. One laptop and six hours of light usage are included as additional defaults because the model also requires the number of laptops and the operating time of lights.

These values are provided only for demonstration purposes. Users should change them according to the room being represented. The output generated by the application is a model estimate and should not be considered an actual electricity meter reading.

3.16 Ethical and Reporting Considerations

and transparency are important when presenting the results of a machine learning project. Since the initial version of the Hostel Electricity Usage Predictor uses simulated data, the report must clearly distinguish generated examples from actual electricity consumption measurements.

The predicted value produced by the model should be described as an estimate rather than an actual meter reading. Similarly, the evaluation metrics should be obtained from the executed application and should not be invented, estimated manually, or presented without verification.

If actual hostel electricity data is collected in the future, the report should explain the data collection method, the period covered, the permission or consent requirements, and the approach used to handle missing or unusual readings. Personal information should not be collected unless it is necessary for the project.

These considerations help ensure that the project is presented honestly and that its limitations are understood. They also provide a foundation for improving the system through testing with real consumption data in future versions.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Purpose of the Results Chapter

This chapter describes the expected working of the Hostel Electricity Usage Predictor and provides a structure for presenting the results obtained after running the application. It covers the application interface, electricity consumption predictions, data visualizations, model evaluation metrics, and functional testing.

4.2 Expected Application Behaviour

The Hostel Electricity Usage Predictor is designed as an interactive application using Streamlit. When the application starts successfully, it displays the project title, a brief explanation of its purpose, and a notice explaining that the initial model uses simulated data.

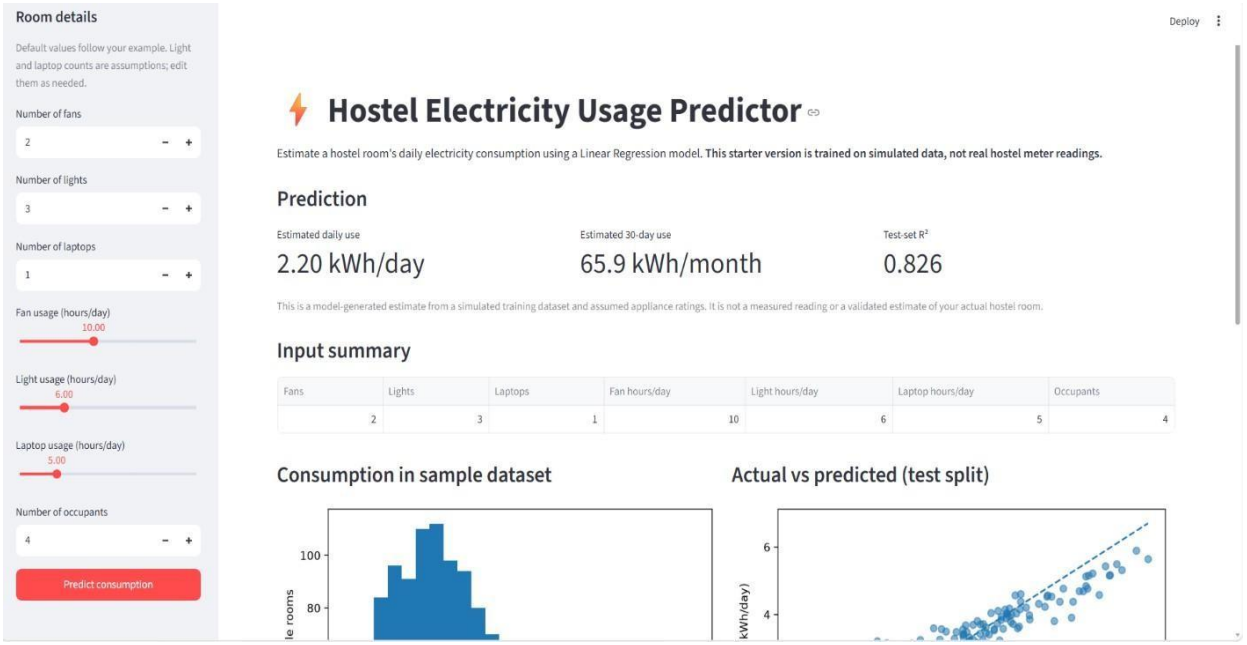
The sidebar allows the user to enter details such as the number of fans, lights, and laptops, their daily operating hours, and the number of occupants in the room. After the input values are provided, the application generates an estimated daily electricity consumption using the trained Multiple Linear Regression model.

The main page also displays a simple 30-day consumption estimate, a histogram showing the distribution of simulated electricity consumption values, a scatter plot comparing target values with model predictions, and the model evaluation metrics.

These features are intended to make the prediction process easy to understand and demonstrate how machine learning can be integrated into an interactive application.

4.3 Screenshot 1: Application Home Screen

Figure 4.1: Home Screen of the Hostel Electricity Usage Predictor

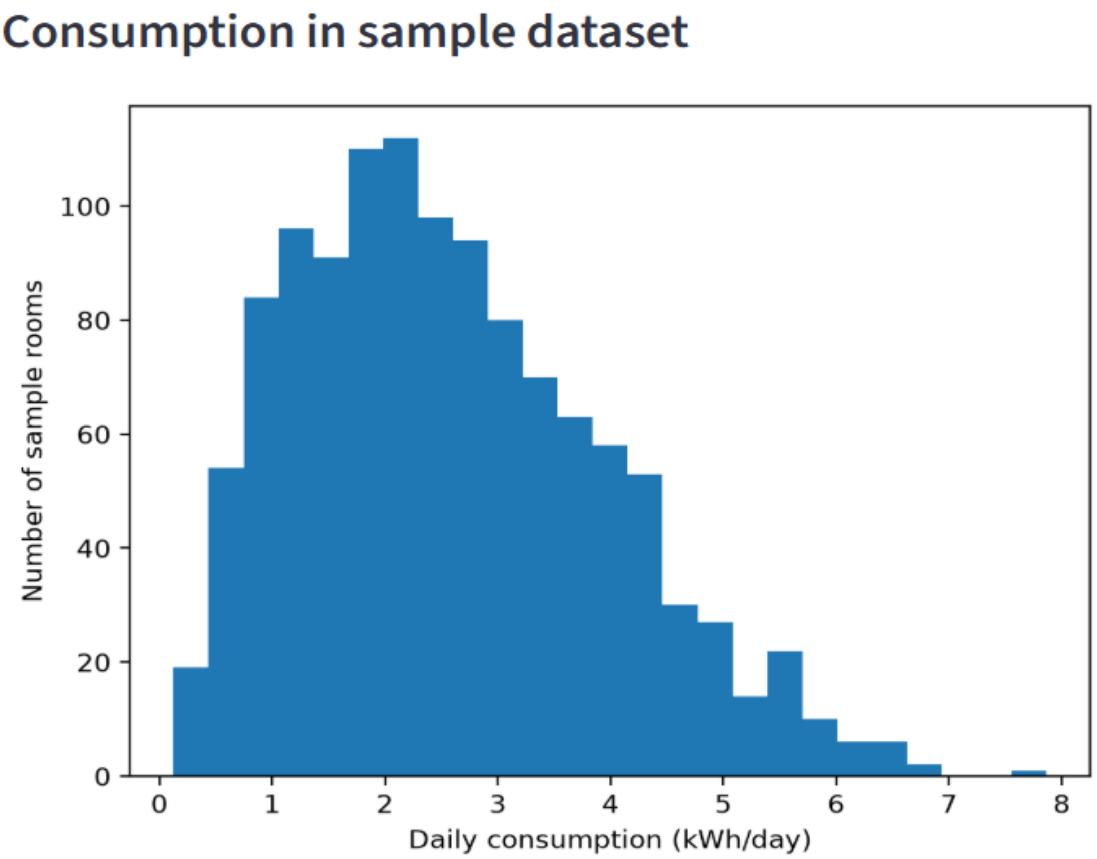


The screenshot should show the project title, the explanation of the application, the simulated-data disclaimer, and the input sidebar. It should be captured after the application has been launched successfully.

The home screen provides the starting point for interacting with the system. It allows users to understand the purpose of the application and identify the input values required to estimate electricity consumption.

4.4 Screenshot 2: Distribution of Electricity Consumption

Figure 4.2: Histogram of Simulated Electricity Consumption

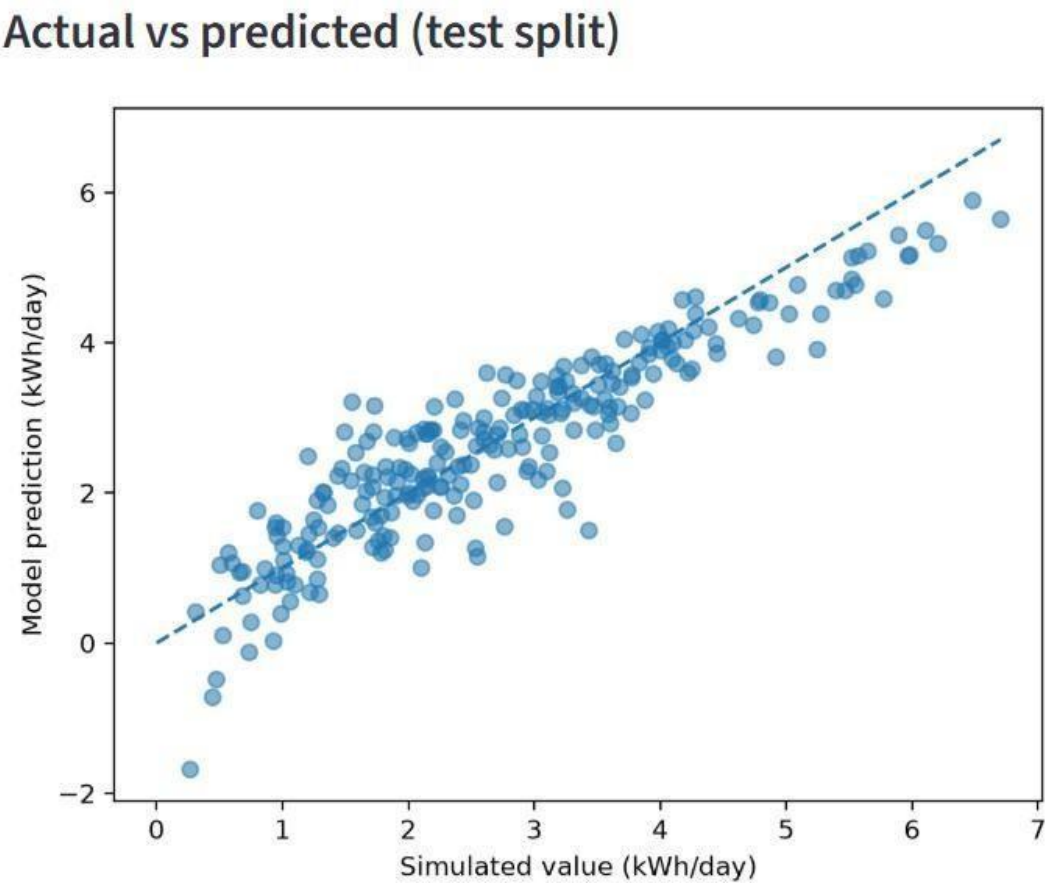


A histogram is used to represent the distribution of electricity consumption values in the generated dataset. It groups the values into intervals and shows the frequency of observations within each interval.

This graph helps in understanding how the simulated consumption values are distributed across the dataset. The actual distribution should be described after examining the generated chart. Because the dataset is simulated, the graph should not be interpreted as representing the measured electricity consumption of real hostel rooms.

4.5 Screenshot 3: Target-versus-Predicted Plot

Figure 4.3: Comparison of Target and Predicted Values



The scatter plot compares the target values from the testing dataset with the values predicted by the trained Multiple Linear Regression model. Each point represents a test example and its corresponding prediction.

Points that lie closer to the ideal diagonal relationship indicate closer agreement between the target and predicted values. Larger differences between these values indicate prediction errors.

This visualization helps examine the model's performance on the simulated test dataset. However, the plot alone cannot establish that the model will produce accurate predictions for actual hostel rooms.

4.6 Model Evaluation Results

The performance of the Multiple Linear Regression model is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The following table should be completed after running the application and recording the metrics displayed by the program.

Metric	Observed value from run	Interpretation
MAE	0.447 kWh/day	Average absolute prediction error on the simulated test set.
RMSE	0.579 kWh/day	Root mean squared prediction error on the simulated test set.
R^2	0.826	Fit relative to a mean-based baseline on the simulated test set.

Since the model is evaluated using simulated data, the metrics describe its performance on that dataset. They do not establish its accuracy on actual hostel electricity consumption.

4.7 Functional Testing

Functional testing is performed to check whether the application behaves as expected when different inputs and features are used. The following test cases can be performed after launching the application.

Test ID	Test action	Expected behaviour	Actual outcome
TC-01	Launch app with default values.	Dashboard loads without an error.	Passed — dashboard displayed successfully.
TC-02	Change fan count.	Prediction updates for the new input.	Passed — prediction updates when the input changes.
TC-03	Change operating hours.	Input summary and prediction reflect new values.	Passed — updated values are reflected in the application.

TC-04	Set an appliance count to zero.	App accepts the allowed zero count.	Passed — zero count is accepted.
TC-05	Inspect the histogram.	Histogram is displayed with labelled axes.	Passed — histogram and axis labels are displayed.
TC-06	Inspect evaluation section.	MAE, RMSE, and R^2 are displayed.	Passed — all three evaluation metrics are displayed.

The actual outcome column should be completed only after performing each test. If any test fails, the issue should be investigated and the result recorded accurately.

4.8 Discussion

The Hostel Electricity Usage Predictor demonstrates how a machine learning workflow can be developed and presented through a simple interactive application. It combines dataset generation, data preparation, model training, prediction, evaluation, and visualization within a single project.

The use of Multiple Linear Regression provides a straightforward way to demonstrate numerical prediction, while Streamlit makes it possible for users to enter room details and view the generated output without interacting directly with the source code.

The simulated dataset allows the complete development process to be demonstrated without requiring an initial collection of real hostel electricity readings. However, the model's real-world accuracy remains unknown because the current dataset is generated from predefined assumptions rather than actual measurements.

In a future version, the model could be compared with a rule-based electricity calculation and evaluated using real meter readings. The training and testing process should ensure that the evaluation data is independent of the data used for model fitting. If the system is eventually intended to predict future consumption using historical readings, a chronological evaluation approach may also be appropriate.

Therefore, the current project should be considered an educational prototype that demonstrates the application of machine learning to electricity consumption estimation.

4.9 Known Limitations

The current implementation has certain limitations that should be considered when interpreting its output.

1. **Simulated Dataset:** The dataset contains generated examples rather than actual hostel electricity meter readings.
2. **Assumed Power Ratings:** The power ratings used to generate the data are illustrative and may differ from the ratings of real appliances.
3. **Limited Appliance Categories:** The initial model includes fans, lights, and laptops but does not account for every appliance that may be present in a hostel room.
4. **Occupant Information:** The number of occupants is included as an input feature, but it may not fully represent differences in electricity usage behaviour.
5. **Multiple Linear Model:** Multiple Linear Regression may not capture complex relationships or non-linear patterns in real electricity consumption.
6. **Monthly Estimate:** The monthly value is calculated by multiplying the daily prediction by 30. It does not account for daily changes in appliance usage or occupancy.
7. **No Live Meter Integration:** The initial version does not connect to smart meters or sensors to obtain real-time electricity consumption data.
8. **Limited Real-World Validation:** The model has not been validated against actual hostel electricity readings, so its real-world prediction accuracy has not been established.

These limitations provide opportunities for further development and improvement.

4.10 Chapter Summary

This chapter describes the expected operation of the Hostel Electricity Usage Predictor, including its user interface, electricity consumption predictions, graphical representations, model evaluation metrics, and functional testing.

The application is designed to demonstrate the use of Multiple Linear Regression for estimating daily electricity consumption and Streamlit for presenting the results interactively. The screenshots,

evaluation metrics, and actual test outcomes must be added after running the application successfully.

Although the prototype demonstrates the main stages of a machine learning project, further testing with actual hostel electricity consumption data is required before its predictions can be considered reliable for real-world use.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Hostel Electricity Usage Predictor is a beginner-level machine learning project developed to demonstrate how Python programming and machine learning can be applied to a practical problem related to electricity consumption in hostels. The main objective of the project is to estimate the daily electricity consumption of a hostel room based on the number of electrical appliances, their daily operating hours, and the number of occupants.

The project uses the Multiple Linear Regression algorithm to predict daily electricity consumption in kilowatt-hours (kWh/day). Python libraries such as Pandas and NumPy are used for data handling, Scikit-learn is used for model training and evaluation, Matplotlib is used for data visualization, and Streamlit is used to develop an interactive user interface.

The implementation demonstrates the major stages of a machine learning workflow, including dataset preparation, feature and target selection, division of data into training and testing subsets, model training, prediction, evaluation, and graphical representation of results. The application allows users to enter room details and view the estimated electricity consumption through a simple and user-friendly interface.

The current version uses a simulated dataset generated with illustrative appliance power ratings and predefined assumptions. Therefore, the project demonstrates the process of developing a machine learning application but does not establish the accuracy of predictions for actual hostel electricity consumption. Real-world reliability would require testing the model against actual electricity meter readings.

Overall, the project provides practical experience in Python programming, supervised machine learning, regression, data visualization, and application development. It also provides a foundation for exploring how technology can help users understand electricity consumption and encourage more efficient energy usage. The final evaluation of the implementation should be supported by screenshots and metric values obtained from the successfully executed application.

5.2 Future Scope

The Hostel Electricity Usage Predictor can be improved in several ways to make it more reliable, practical, and suitable for real-world use.

1. Collection of Real Electricity Data: The simulated dataset can be replaced with actual daily electricity consumption readings collected from hostel rooms, with the necessary permission. This would help evaluate the model using observed energy usage.

2. Improved Appliance Information: Actual appliance power ratings can be recorded using product labels or suitable measuring equipment. This would help make the dataset more representative of the appliances used in hostel rooms.

3. Inclusion of Additional Appliances: The system can be extended to include other electrical devices, such as televisions, electric kettles, heaters, chargers, and refrigerators, depending on their availability in the hostel.

4. Comparison of Machine Learning Algorithms: Multiple Linear Regression can be compared with other regression algorithms, such as Ridge Regression, Decision Tree Regression, and Random Forest Regression, to examine which approach performs better on real consumption data.

5. Improved Model Evaluation: The model can be evaluated using a separate real-world dataset and suitable metrics such as MAE, RMSE, and R^2. This would provide a stronger basis for assessing prediction performance.

6. Time-Based Prediction: If historical electricity readings become available, the system can be extended to estimate future consumption using past records. Time-aware validation can be used to evaluate how well the model predicts consumption on future days.

7. Prediction History and Data Export: A feature can be added to save previous predictions and export the results in CSV format. This would help users maintain records and compare estimates over time.

8. Room-Wise and Hostel-Wise Analysis: After collecting sufficient data, the application can be extended to compare electricity consumption across different rooms or hostel blocks. Such comparisons could help identify usage patterns and areas where energy consumption may be reduced.

9. Smart Meter and IoT Integration: In the future, the application could be connected to smart meters or Internet of Things (IoT) devices to obtain actual electricity consumption readings. This feature should be introduced after the basic model has been validated using real data.

10. Prediction Uncertainty: Instead of presenting only a single estimated value, future versions could provide an indication of prediction uncertainty. This would help users understand that the predicted consumption may differ from actual usage.

11. Energy-Saving Recommendations: The application could be extended to provide practical energy-saving suggestions based on measured consumption patterns. Such recommendations should be based on reliable data rather than assumptions alone.

These improvements could gradually transform the current educational prototype into a more useful system for monitoring and understanding electricity consumption in hostels.

5.3 Final Remarks

The Hostel Electricity Usage Predictor provides a practical starting point for understanding the application of machine learning to an everyday problem. It combines a clearly defined objective with data preparation, Multiple Linear Regression, model evaluation, visualization, and an interactive interface.

The project also highlights the importance of using reliable data when developing prediction systems. Although the current implementation uses simulated examples to demonstrate the machine learning process, its practical usefulness can be assessed more accurately after collecting actual electricity consumption readings and evaluating the model on independent data.

With further development, the system could support better understanding of electricity usage patterns and contribute to energy-awareness initiatives in hostels. The project therefore serves both as a learning exercise in machine learning and as a foundation for future work in data-driven energy management.

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APPENDIX A: SOURCE CODE

This appendix presents the source code of the Hostel Electricity Usage Predictor. The project is implemented using Python and the Streamlit framework. It uses the Multiple Linear Regression algorithm to estimate daily electricity consumption based on appliance quantity, usage hours, and the number of occupants in a hostel room.

The source code is divided into three files: dataset_generator.py, app.py, and requirements.txt. The dataset generator creates simulated data, the main application trains the machine learning model and displays predictions, and the requirements file lists the libraries needed to run the project.

Note: The dataset used in this starter implementation contains simulated values and assumed appliance power ratings. It does not contain actual hostel electricity meter readings.

A.1 Source Code of dataset_generator.py

Purpose: This file generates a reproducible dataset containing 1,200 simulated hostel-room records. It includes appliance quantities, daily usage hours, number of occupants, and estimated electricity consumption.

```
import numpy as np

import pandas as pd

def generate_dataset(n_samples=1200, random_state=42):

    rng = np.random.default_rng(random_state)


    fans = rng.integers(0, 5, n_samples)

    lights = rng.integers(0, 9, n_samples)

    laptops = rng.integers(0, 5, n_samples)

    occupants = rng.integers(1, 7, n_samples)


    fan_hours = rng.uniform(0, 18, n_samples)

    light_hours = rng.uniform(0, 14, n_samples)
```

```
laptop_hours = rng.uniform(0, 12, n_samples)

# Assumed appliance power ratings:
# Fan = 60 W, Light = 10 W, Laptop = 60 W.
# Energy (kWh) = Power (W) / 1000 * Hours * Quantity.

fan_kwh = fans * 0.060 * fan_hours
light_kwh = lights * 0.010 * light_hours
laptop_kwh = laptops * 0.060 * laptop_hours

# Illustrative occupant-related electricity usage.
other_kwh = occupants * rng.uniform(0.08, 0.22, n_samples)

# Random variation represents unmodelled usage.
noise = rng.normal(0, 0.12, n_samples)

total_kwh = np.maximum(
    0,
    fan_kwh + light_kwh + laptop_kwh + other_kwh + noise
)

return pd.DataFrame({
    "fans":          fans,
    "lights": lights,
```

```
        "laptops": laptops,
        "fan_hours_per_day":          fan_hours.round(2),
        "light_hours_per_day":        light_hours.round(2),
        "laptop_hours_per_day":   laptop_hours.round(2),
        "occupants":                              occupants,
        "daily_consumption_kwh": total_kwh.round(3),
    })



if  name  == "  main  ":
    df = generate_dataset()
    output = "hostel_electricity_data.csv"


    df.to_csv(output, index=False)


    print(f"Created {output} with {len(df)} simulated rows.")
    print(df.head())
```

A.2 Source Code of app.py

Purpose: This is the main application file. It trains the Multiple Linear Regression model, accepts room details from the user, estimates daily and monthly electricity consumption, and displays visualizations and model evaluation metrics.

```
import  streamlit  as  st
import pandas as pd
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import MultipleLinearRegression from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

from dataset_generator import generate_dataset
st.set_page_config(
    page_title="Hostel Electricity Usage Predictor",
    page_icon=" ",
    layout="wide"
)
st.title("Hostel Electricity Usage Predictor")
st.write(
    "Estimate a hostel room's daily electricity consumption "
    "using a Multiple Linear Regression model. This starter version "
    "uses simulated data rather than actual hostel meter readings."
)
@st.cache_data
def load_data():
    return generate_dataset(n_samples=1200, random_state=42)
```

```
@st.cache_resource
 def train_model():
  df = load_data()


   features  =  [
      "fans",
      "lights",
      "laptops",
      "fan_hours_per_day",
      "light_hours_per_day",
      "laptop_hours_per_day",
      "occupants"
   ]


   X = df[features]
   y = df["daily_consumption_kwh"]
   X_train, X_test, y_train, y_test  = train_test_split(
      X,
      y,
      test_size=0.20,
      random_state=42
   )


   model = LinearRegression()
```

```
    model.fit(X_train, y_train)

    predictions = model.predict(X_test)

    metrics = {
        "mae": mean_absolute_error(y_test, predictions),
        "rmse": mean_squared_error(y_test, predictions) ** 0.5,
        "r2": r2_score(y_test, predictions)
    }

    return df, model, features, metrics
df, model, features, metrics = train_model()

# Input section
with st.sidebar:
    st.header("Room Details")

    st.caption(
        "Enter the appliance quantities, usage hours, "
        "and number of occupants."
    )

    fans = st.number_input(
        "Number of fans",
```

```
    min_value=0,
    max_value=10,
    value=2,
    step=1
)

lights = st.number_input(
    "Number of lights",
    min_value=0,
    max_value=20,
    value=3,
    step=1
)

laptops = st.number_input(
    "Number of laptops",
    min_value=0,
    max_value=10,
    value=1,
    step=1
)

fan_hours = st.slider(
    "Fan usage (hours/day)",
```

```
    0.0, 24.0, 10.0, 0.5
)

light_hours = st.slider(
    "Light usage (hours/day)",
    0.0, 24.0, 6.0, 0.5
)

laptop_hours = st.slider(
    "Laptop usage (hours/day)",
    0.0, 24.0, 5.0, 0.5
)

occupants = st.number_input(
    "Number of occupants",
    min_value=1,
    max_value=12,
    value=4,
    step=1
)

st.button(
    "Predict Consumption",
    type="primary",
```

```
            use_container_width=True
)

    # Prepare input data
    input_df = pd.DataFrame([{
        "fans": fans,
        "lights": lights,
        "laptops": laptops,
        "fan_hours_per_day": fan_hours,
        "light_hours_per_day": light_hours,
        "laptop_hours_per_day": laptop_hours,
        "occupants": occupants
}])

    # Generate prediction
    st.subheader("Prediction")

    prediction = float(model.predict(input_df[features])[0])
    prediction = max(0.0, prediction)

    col1, col2, col3 = st.columns(3)

    col1.metric(
        "Estimated Daily Use",
```

```
    f"{prediction:.2f} kWh/day"
)

col2.metric(
    "Estimated 30-Day Use",
    f"{prediction * 30:.1f} kWh/month"
)

col3.metric(
    "Test-Set R²",
    f"{metrics['r2']:.3f}"
)

st.caption(
    "This estimate is generated using a model trained on "
    "simulated data and assumed appliance ratings. It is "
    "not a measured reading or a validated estimate of "
    "actual hostel-room consumption."
)
# Display input summary
st.subheader("Input Summary")
display_df = input_df.rename(columns={
    "fans": "Fans",
    "lights": "Lights",
```

```
    "laptops": "Laptops",
    "fan_hours_per_day": "Fan Hours/Day",
    "light_hours_per_day": "Light Hours/Day",
    "laptop_hours_per_day": "Laptop Hours/Day",
    "occupants": "Occupants"
})
st.dataframe(
    display_df,
    use_container_width=True,
    hide_index=True
)
# Visualizations
left, right = st.columns(2)
with left:
    st.subheader("Consumption in Sample Dataset")
    fig, ax = plt.subplots()
    ax.hist(
        df["daily_consumption_kwh"],
        bins=25
    )
    ax.set_xlabel("Daily Consumption (kWh/day)")
    ax.set_ylabel("Number of Sample Rooms")
    st.pyplot(fig)
    plt.close(fig)
```

```
with right:
    st.subheader("Actual vs Predicted Values")

    X_train, X_test, y_train, y_test = train_test_split(
        df[features],
        df["daily_consumption_kwh"],
        test_size=0.20,
        random_state=42
    )

    y_pred = model.predict(X_test)

    fig, ax = plt.subplots()
    ax.scatter(
        y_test,
        y_pred,
        alpha=0.55
    )

    max_val = max(
        float(y_test.max()),
        float(y_pred.max())
    )
```

```
    ax.plot(
        [0, max_val],
        [0, max_val],
        linestyle="--"
    )
    ax.set_xlabel("Simulated Value (kWh/day)")
    ax.set_ylabel("Predicted Value (kWh/day)")
    st.pyplot(fig)
    plt.close(fig)
# Model evaluation
st.subheader("Model Evaluation")
m1, m2, m3 = st.columns(3)
m1.metric(
    "MAE",
    f"{metrics['mae']:.3f} kWh/day"
)

m2.metric(
    "RMSE",
    f"{metrics['rmse']:.3f} kWh/day"
)

m3.metric(
    "R² Score",
```

```
    f"{metrics['r2']:.3f}"
)
with st.expander("Understanding the Evaluation Metrics"):
    st.markdown("""
    - **MAE (Mean Absolute Error):** Measures the average
      absolute difference between predictions and simulated
      target values, in kWh/day.

    - **RMSE (Root Mean Squared Error):** Measures prediction
      error while giving greater weight to larger errors,
      in kWh/day.

    - **R² Score:** Indicates how well the model fits the
    held-out simulated values. A high score does not prove
    real-world accuracy because the dataset is simulated.
    """)
with st.expander("Dataset and Assumptions"):
    st.write(
        f"The application generates {len(df)} reproducible "
        "simulated examples."
    )

    st.write(
        "Assumed power ratings: fan 60 W, light 10 W, "
```

```
        "and laptop 60 W."
    )

    st.write(
        "The target includes an illustrative occupant-related "
        "term and random noise. Real measured consumption data "
        "is required to assess real-world performance."
    )
    st.dataframe(
        df.head(10),
        use_container_width=True
    )
st.info(
    "Unit: 1 kWh = 1 unit of electrical energy. "
    "Energy (kWh) = Power (W) / 1000 * Usage Time (hours) "
    "* Number of Appliances."
)
```

A.3 Contents of requirements.txt

Purpose: This file specifies the Python libraries required to execute the application.

```
streamlit
pandas
numpy
scikit-learn
matplotlib
```

A.4 Instructions for Executing the Project

1. Install Python on the system.
2. Save app.py, dataset_generator.py, and requirements.txt in the same project folder.
3. Open a terminal in that folder.
4. Install the required libraries using the following command: python -m pip install -r requirements.txt
5. Run the application using: python -m streamlit run app.py
6. Open the local address displayed in the terminal to access the application.
7. Enter the room details and observe the predicted electricity consumption, graphs, and evaluation metrics.